

JUNJIE LI

@ alvinbluy@gmail.com |  Portfolio |  LinkedIn |  GitHub |  Bothell, WA, Open to Relocation

EDUCATION

Northeastern University

Seattle, WA | Sep 2023 - Dec 2025

Master of Science in Computer Science (GPA: 3.9/4.0)

Teaching Assistant: Object Oriented Design (Spring 2024)

Temple University

Philadelphia, PA

M.S. in Business Research (Quantitative Marketing), PhD quit

Nankai University

Tianjin, China

B.S. in Mathematics (Chern Honor Class)

EXPERIENCE

Berkshire Hathaway HomeServices: Capstone Intern

Seattle, WA | Jun 2024 - Aug 2024

Machine Learning Engineer

- Maintained data pipelines using Apache Airflow, ensuring seamless data flow.
- Customized and fine-tuned deep learning models (N-Beats series) for housing price and inventory predictions, improving model accuracy by 30%. Applied Agile principles to iteratively refine data acquisition processes and update models, resulting in a 25% reduction in development cycle time.

PhD Research: Understand User's Video Appreciation using Live Comments

Philadelphia, PA

Analyzed live comments with data mining, computer vision and natural language processing techniques.

- Preprocessed live comment data with Pandas and NumPy, and video content using OpenCV and BLIP. Extracted features from frames and text with ResNet-50 and pre-trained BERT embeddings.
- Improved data processing scalability and efficiency by 5x using Dask and NVIDIA RAPIDS suite.

PROJECTS

Search Engine for Emojis | [Link](#)

LLM-RAG Python/Go Web

"Emoji Search", a Llama 2-powered search engine finding emojis based on web services.

- Built precise search feature with Retrieval-Augmented Generation (RAG). Achieved low latency (< 200ms) at high volumes (10,000 queries/second) using Amazon OpenSearch Serverless and AWS Bedrock services.
- Achieved smoother search experience by adopting dynamic microservice discovery with Consul and balancing queries with NGINX, resulting in 30% fewer service failures and 15% faster response times.
- Implemented real-time search query analytics with Kafka for scalability and more insights. Containerized the app in Docker and deployed it on GCP Kubernetes Engine via GitHub Actions CI/CD pipeline for efficiency.

AsyncTask Scheduler | Collaborated with [Hui Zhou](#) | [Link](#)

Distributed Go Framework

"Async4Ai", a Go framework to asynchronously schedule tasks for parallel data processing.

- Decoupled a task-manager agent of RESTful API from workers in object-oriented design. Rewrote priority scheduling and flexible retry strategies, aligning with scalable system design best practices.
- Designed loosely coupled table normalization & sharding logic in MySQL.
- Performed stress testing and improved performance from 500 QPS to 2000 QPS by using distributed locks and caching with Redis and reconfiguring the database connection pool.

Anonymous Academic Career Hub | [Link](#)

Full-stack JavaScript Web

"Business Job Market Rumors", a discussion board for academic jobs in business schools.

- Utilized Express/Node as backend and React as frontend in a client-server model, integrating Tailwind CSS.
- Implemented role-based access control, context-based authentication using JWT, and automated content moderation using Flask to ensure a secure and respectful discussion environment.
- Deployed on Azure App Service with MongoDB hosted on Azure Cosmos DB.

TECHNICAL SKILLS

Languages & Databases: Python(5yr), Java(2yr), Golang(2yr), C/C++(1yr), JavaScript, C, SQL; MongoDB, Redis

Frameworks: CUDA, Spark, Dask, PyTorch, TensorFlow, Langchain, Streamlit, Express, Node, Angular, FastAPI

Tools: Dataiku, NeMo, Flask, Gradle, JUnit, Postman, GitHub Actions, Docker, Spark, Kubernetes

Miscellaneous: Linux, AWS, Azure ([AI Engineer Associate Certification](#)), GCP